NightfallLX (v2.0) — Complete Technical Manual & Property Reference

Next-Generation High-Performance Web Assessment, Vulnerability Scanner & Deep OSINT Recon Suite

Runtime: Pure Go (Golang) 1.21+ • **Zero Dependencies** • **Target Audience:** White Hat Hackers, Pentesters & Red Teams

1. Executive Architecture Overview

NightfallLX is an evidence-first security platform engineered for high-throughput automated vulnerability assessment, passive OSINT reconnaissance, and nanosecond concurrency testing. It features 15 specialized packages: deep passive OSINT (Certificate Transparency via crt.sh, DNS SPF/DMARC email spoofing audit, CNAME Subdomain Takeover detection, and Wayback Machine historical archive discovery), intelligent user enumeration with differential auto-regex inference, TruffleHog-grade secrets scanning, a Turbo Intruder-style nanosecond gate-sync race engine, Nuclei v3 YAML template execution, multi-threaded port scanning with NIST NVD CVE correlation, WAF adaptive rate throttling, and multi-format reporting (Interactive Offline Cyber HTML Dashboard, SARIF v2.1.0, and JSON).

2. Exhaustive Properties & CLI Flags Reference (Complete 3-Column Table)

Every command-line flag, configuration property, and operational mode supported by NightfallLX (Go Edition v2.0):

Property / Flag	Description / Purpose	Example Usage
-u	Target base URL or URL template with injection marker (FUZZ or *). Required for web scanning and fuzzing unless -r, -domain, or -scan is provided.	`nightfalllx -u "http://target.lab/?id=1"` `nightfalllx -u "http://target.lab/FUZZ"`
-r	Burp Suite raw HTTP request file export. Parses HTTP method, full URL, headers, cookies, query/POST parameters, and injection point markers (*).	`nightfalllx -r request.txt -m all`
-domain	Target root domain for Subdomain/VHost discovery, OSINT intelligence, or port scanning.	`nightfalllx -domain target.com -osint` `nightfalllx -sub -domain target.htb -u http://10.129.20.10`
-https	Forces HTTPS protocol scheme for the target URL if not explicitly included in the address string.	`nightfalllx -u target.lab -https`
-ssl	Alias for -https. Enforces TLS encrypted communication.	`nightfalllx -u target.lab -ssl`
-scope	Restricts all requests strictly to authorized domains. Accepts comma-separated list or path to scope.txt. Refuses off-scope hosts.	`nightfalllx -u http://target.lab -scope "target.lab,*.target.lab"` `nightfalllx -r req.txt -scope scope.txt`
-scope-off	Explicitly disables scope checking. Bypasses domain validation (use only when permitted).	`nightfalllx -u http://target.lab -scope-off`

Property / Flag	Description / Purpose	Example Usage
-b2	Secondary cookie value for automated Privilege Escalation, BOLA, and IDOR testing between two user sessions.	`nightfalllx -u "http://target.lab/account" -b "session=user1" -b2 "session=user2" -m idor`
-H2	Secondary custom authorization header for IDOR/RBAC testing (repeatable for multiple headers).	`nightfalllx -u "http://target.lab/api" -H "Auth: Bearer tok1" -H2 "Auth: Bearer tok2" -m idor`
-osint	Executes comprehensive passive OSINT: crt.sh Certificate Transparency, DNS records, SPF/DMARC email spoofing audit, CNAME takeover, and Wayback Machine archive discovery.	`nightfalllx -osint -domain example.com` `nightfalllx -osint -u https://target.lab -timeout 25`
-shodan	Runs Shodan OSINT reconnaissance: queries public IP allocations, open ports, software banners, hostnames, and known CVEs.	`nightfalllx -shodan -u http://target.lab`
-shodan-key	Custom Shodan API key. Overrides built-in defaults or the SHODAN_API_KEY environment variable.	`nightfalllx -shodan -u http://target.lab -shodan-key "KEY"`
-tech	Deep Wappalyzer-grade technology stack and exact version fingerprinting (Web Server, PHP, Frameworks, Template Engines, Security Headers).	`nightfalllx -tech -u http://target.lab`
-map	Spiders target web architecture recursively and produces a sitemap. Extracts endpoints from JS files and parses robots.txt / sitemap.xml.	`nightfalllx -map -u http://target.lab -depth 3 -max-urls 100`
-spider	Alias for -map. Executes recursive web crawler.	`nightfalllx -spider -u http://target.lab`
-crawl	Alias for -map. Executes recursive web crawler.	`nightfalllx -crawl -u http://target.lab`
-depth	Maximum recursion depth for the web crawler (default: 3).	`nightfalllx -map -u http://target.lab -depth 4`
-max-urls	Maximum number of unique URLs to discover during spidering before termination (default: 50).	`nightfalllx -map -u http://target.lab -max-urls 150`
-mine-params	Runs PortSwigger-style hidden parameter miner. Batches candidates (25/req) with canary tokens to identify unlinked parameters via reflection, status, and length anomalies.	`nightfalllx -mine-params -u "http://target.lab/index.php" -X POST`
-param-miner	Alias for -mine-params.	`nightfalllx -param-miner -u http://target.lab/search`

-m	Comma-separated list of vulnerability testing modules to run, or 'all' (default: 'all'). Available: sqli, ssti, lfi, cmdi, xss, cors, redirect, ssrf, jwt, waf, idor, csrf, xxe, nosqli, upload, headers, exposure, hostheader, crlf, secrets, race, userenum.	`nightfalllx -u "http://target.lab/?id=1" -m "sqli,xss,lfi"`
		`nightfalllx -u "http://target.lab/login" -m userenum`
		`nightfalllx -u "http://target.lab" -m secrets`

-scan	Enables multi-threaded TCP network port scanning mode. Probes target host ports, extracts service banners, and queries vulnerabilities.	`nightfalllx -scan -domain target.lab -scan-ports "80,443,8080"`
-scan-ports	Port range or profile for network scanner: 'top100' (default), ranges like '1-1024', or comma-delimited lists like '80,443,135,445'.	`nightfalllx -scan -domain 192.168.1.10 -scan-ports "1-1000"` `nightfalllx -scan -domain target.com -scan-ports "80,443"`

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-scan-cve	Enriches discovered open ports and service version banners with live queries against the NIST NVD CVE vulnerability database.	`nightfalllx -scan -domain target.lab -scan-ports "top100" -scan-cve`
-race	Executes nanosecond gate-synchronized race condition and limit bypass attack (Turbo Intruder style). Releases burst requests simultaneously.	`nightfalllx -u "http://target.lab/api/redeem" -X POST -d '{"code":"VIP"}' -race -race-threads 30`
-race-threads	Number of concurrent parallel burst requests dispatched per round during race condition attack (default: 20).	`nightfalllx -u "http://target.lab/cart/checkout" -race -race-threads 40`
-race-rounds	Number of burst rounds to repeat during race testing (default: 2).	`nightfalllx -u "http://target.lab/vote" -race -race-rounds 3`
-templates	Path to a single Nuclei YAML template file or directory of templates to parse and execute against the target.	`nightfalllx -u "http://target.lab" -templates ./cves/CVE-2023-1234.yaml` `nightfalllx -u "http://target.lab" -templates ./templates/`
-template-tags	Comma-separated tag filter for Nuclei templates when loading from a directory (e.g. 'cve,misconfig,auth').	`nightfalllx -u "http://target.lab" -templates ./templates -template-tags "cve,critical"`
-X	HTTP Request Method: GET, POST, PUT, DELETE, PATCH, OPTIONS, HEAD (default: 'GET').	`nightfalllx -u "http://target.lab/api" -X POST -d "action=test"`
-d	POST / PUT body data string. Supports urlencoded parameters, JSON payloads, or injection marker FUZZ.	`nightfalllx -u "http://target.lab/login" -X POST -d "user=admin&pass;=FUZZ"`
-H	Custom HTTP Header to include in requests. Repeatable flag for passing multiple headers.	`nightfalllx -u http://target.lab -H "Authorization: Bearer SECRET" -H "X-Custom: 1"`
-b	HTTP Cookie header value to attach to all requests.	`nightfalllx -u http://target.lab -b "PHPSESSID=d92j10dj12; role=admin"`
-p	HTTP proxy URL for forwarding outbound scanner traffic (e.g. Burp Suite, OWASP ZAP, or Caido).	`nightfalllx -u http://target.lab -p http://127.0.0.1:8080`
-rate	Token-bucket rate limiter specifying maximum requests per second. Default 0 (unlimited). Prevents server overload or DoS.	`nightfalllx -u http://target.lab/FUZZ -w wordlist.txt -rate 50`
-retry	Maximum automatic retry attempts when encountering HTTP 429 (Too Many Requests), HTTP 503, or connection dropouts (default: 0).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -retry 3`
-retry-delay	Base exponential backoff delay in milliseconds between retries (default: 500). Dynamically respects Retry-After headers.	`nightfalllx -u http://target.lab -retry 3 -retry-delay 1000`
-http2	Enables or disables HTTP/2 transport protocol support over TLS (default: true).	`nightfalllx -u https://target.lab -http2=false`
-raw	Bypasses Go's HTTP client and header normalization by transmitting verbatim bytes over direct raw TCP/TLS socket connections.	`nightfalllx -u http://target.lab/FUZZ -w list.txt -raw`

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-tamper-headers	Injects WAF bypass headers (X-Originating-IP, X-Forwarded-For, X-Remote-IP, X-Custom-IP-Authorization) to circumvent IP-based access controls.	`nightfalllx -u http://target.lab/admin -tamper-headers`
-chunked	Transmits request body payloads using Transfer-Encoding: chunked to evade payload inspection firewalls.	`nightfalllx -u http://target.lab/upload -X POST -d "payload" -chunked`
-adaptive	Activates intelligent WAF throttling: automatically slows concurrency upon 429/403 responses, injects jitter, and rotates browser fingerprints.	`nightfalllx -u http://target.lab/FUZZ -w list.txt -adaptive -rotate-ua`
-jitter	Adds a randomized sleep delay between 0 and N milliseconds between adaptive fuzz requests to randomize traffic cadence (default: 0).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -adaptive -jitter 200`
-rotate-ua	Dynamically cycles through realistic browser User-Agent fingerprints (Chrome, Firefox, Safari, Edge across Windows, macOS, Linux).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -rotate-ua`
-adaptive-min	Concurrency floor for adaptive rate throttling; prevents worker thread starvation (default: 1).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -adaptive -adaptive-min 2`
-k	Insecure TLS verification bypass (default: true). Allows auditing laboratory targets with self-signed, untrusted, or expired certificates.	`nightfalllx -u https://192.168.1.50 -k`
-timeout	HTTP network timeout duration in seconds per request (default: 15).	`nightfalllx -u http://target.lab -timeout 30`
-o	Output filename base prefix for generated reports (default: 'lab_report'). Generates .md, .json, etc.	`nightfalllx -u http://target.lab -m all -o audit_q3`
-report-html	Path to generate an interactive offline Cyber Dark Mode HTML Dashboard report with filterable tables and ready-to-run curl reproduction commands.	`nightfalllx -u http://target.lab -m all -report-html ./reports/dashboard.html`
-report-sarif	Path to output a standardized SARIF v2.1.0 vulnerability report suitable for direct ingestion into GitHub Code Scanning, GitLab CI, or DefectDojo.	`nightfalllx -u http://target.lab -m all -report-sarif ./reports/results.sarif`
-report-json	Path to output a structured JSON report containing findings, CVSS scores, remediation guidance, and evidence payloads.	`nightfalllx -u http://target.lab -m all -report-json ./reports/report.json`
-w	Path to a wordlist file on disk for directory, parameter, or virtual host fuzzing.	`nightfalllx -u http://target.lab/FUZZ -w /usr/share/wordlists/dirb/common.txt`
-t	Number of concurrent worker goroutines for fuzzing, spidering, or crawling (default: 40).	`nightfalllx -u http://target.lab/FUZZ -w wordlist.txt -t 80`
-fuzz	Explicitly enables high-speed fuzzing mode without requiring the FUZZ keyword in the URL.	`nightfalllx -u http://target.lab/ -w wordlist.txt -fuzz`
-sub	Enables virtual host (VHost) and subdomain discovery mode (ffuf-compatible). Fuzzes HTTP Host headers against target IP.	`nightfalllx -sub -domain target.lab -u http://10.129.20.10 -w subdomains.txt`
-subdomain	Alias for -sub. Activates VHost discovery mode.	`nightfalllx -subdomain -domain target.lab -u http://10.129.20.10`

Property / Flag	Description / Purpose	Example Usage
-ac	Auto-calibrates baseline wildcard / catch-all responses before fuzzing (default: true). Eliminates false positives by discarding baseline signatures.	`nightfalllx -u http://target.lab/FUZZ -w wordlist.txt -ac=true`
-mc	Match HTTP status codes (comma-separated). Only responses with these codes are reported.	`nightfalllx -u http://target.lab/FUZZ -w list.txt -mc "200,204,301,302,403"`
-fc	Filter HTTP status codes (comma-separated, default: '404'). Responses matching these codes are discarded.	`nightfalllx -u http://target.lab/FUZZ -w list.txt -fc "404,500"`
-ms	Match response sizes in bytes (comma-separated).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -ms "1024,2048"`
-fs	Filter response sizes in bytes (comma-separated). Discards fixed-length error responses.	`nightfalllx -u http://target.lab/FUZZ -w list.txt -fs "0,421"`
-mw	Match response word counts (comma-separated).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -mw "15,22"`
-fw	Filter response word counts (comma-separated).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -fw "45"`
-ml	Match response line counts (comma-separated).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -ml "10,20"`
-fl	Filter response line counts (comma-separated).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -fl "2"`
-mr	Match response body regular expression. Only responses matching the regex pattern are recorded.	`nightfalllx -u http://target.lab/FUZZ -w list.txt -mr "(?i)admin|dashboard"`
-fr	Filter response body regular expression. Discards responses matching the regex pattern (e.g. custom error messages).	`nightfalllx -u http://target.lab/FUZZ -w list.txt -fr "(?i)page not found|access denied"`
-auto-regex	Automatically infers differential error regexes from baseline canary responses and applies them as dynamic filters to eliminate soft-404 false positives.	`nightfalllx -u http://target.lab/FUZZ -w list.txt -auto-regex`
-i	Launches the interactive mouse-controlled Terminal User Interface (TUI) dashboard with full ANSI SGR mouse navigation and neon cyber visualizers.	`nightfalllx -i`
-ui	Alias for -i. Launches interactive mouse TUI dashboard.	`nightfalllx -ui`
-tui	Alias for -i. Launches interactive mouse TUI dashboard.	`nightfalllx -tui`
-doctor	Pre-flight connectivity diagnostic suite: verifies DNS resolution, TCP handshakes, TLS 1.3 protocol versions, cipher suites, proxy routing, and system CPU health.	`nightfalllx -doctor -u https://target.lab -p http://127.0.0.1:8080`
-troubleshoot	Alias for -doctor. Runs pre-flight diagnostics.	`nightfalllx -troubleshoot -u http://target.lab`
-version	Displays current program version, build timestamp, and runtime architecture.	`nightfalllx -version`

Property / Flag	Description / Purpose	Example Usage
-V	Alias for -version.	`nightfalllx -V`

3. Operational Workflows & Attack Scenarios

Workflow A: Zero-Knowledge Passive OSINT Reconnaissance

Engage a target domain without emitting intrusive traffic:
Command: nightfalllx -osint -domain example.com -timeout 20
Features: Extracts SSL certificates via crt.sh, queries DNS records, verifies SPF/DMARC policies for email spoofing vulnerability, detects Subdomain Takeover on dangling CNAMEs, and uncovers historical sensitive endpoints (.env, .sql, .bak) from the Wayback Machine.

Workflow B: High-Speed Web Fuzzing with Auto-Regex Filtering

Eliminate soft-404 false positives on applications returning HTTP 200 for error pages:
Command: nightfalllx -u "http://target.lab/FUZZ" -w /usr/share/wordlists/dirb/common.txt -t 50 -auto-regex
Mechanism: Dispatches canary tokens during calibration, synthesizes dynamic regex filters for error strings, and automatically drops non-existent hits without manual intervention.

Workflow C: User Enumeration on Login Endpoints

Detect registered user accounts through differential response and execution timing analysis:
Command: nightfalllx -u "http://target.lab/login" -X POST -d "username=admin&password;=Password123" -m userenum
Mechanism: Compares canary baseline responses against candidate accounts. Flags differences in error signatures, status codes, response lengths, or bcrypt/argon2 execution latency (>180ms).

Workflow D: Limit Bypass via Turbo Race Condition

Test concurrency flaws and race condition vulnerabilities in financial checkouts, coupon redemption, or voting:
Command: nightfalllx -u "http://target.lab/api/apply-coupon" -X POST -d '{"coupon":"PROMO50"}' -race -race-threads 30 -race-rounds 2
Mechanism: Pre-constructs requests in memory and releases 30 goroutines simultaneously via channel gate synchronization. Analyzes status code collisions and discrete body variants while filtering dynamic nonces.

Workflow E: Full Enterprise Vulnerability Assessment & Reporting

Execute all 22 assessment modules through a proxy and generate interactive dashboards:
Command: nightfalllx -u "http://target.lab/app.php?id=1" -m all -p http://127.0.0.1:8080 -report-html ./reports/audit.html -report-sarif ./reports/audit.sarif
Mechanism: Tests SQLi, SSTI, LFI, CMDi, XSS, CORS, SSRF, JWT, WAF, IDOR, CSRF, XXE, NoSQLi, Upload, Headers, Exposure, HostHeader, CRLF, Secrets, Race, and UserEnum. Produces an offline Cyber Dark Mode HTML Dashboard and standard SARIF v2.1.0.

4. Pre-Flight Connectivity & Diagnostics (-doctor)

Run the automated diagnostic subsystem before conducting assessments to isolate bottlenecks:
Command: nightfalllx -doctor -u https://target.lab -p http://127.0.0.1:8080
• **DNS Resolution:** Verifies resolver availability and IP resolution.
• **TCP Handshake:** Measures latency and connection availability.
• **TLS Handshake:** Audits TLS version (TLS 1.2 vs 1.3) and negotiated cipher suites.
• **Proxy Connectivity:** Confirms upstream HTTP proxy status and response integrity.

Security & Legal Notice: NightfallLX is engineered exclusively for authorized security audits, sanctioned penetration tests, and educational vulnerability research. Executing security testing against target infrastructures without prior explicit written authorization is illegal.